

Lesson 12: Composite & Inverse Functions — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. $f(x)=x+4$, $g(x)=2x$. Find $(f \circ g)(x)$.

2. $f(x)=3x$, $g(x)=x-1$. Find $(f \circ g)(x)$.

3. $f(x)=x^2$, $g(x)=x+1$. Find $(f \circ g)(2)$.

4. $f(x)=x+5$. Find $f^{-1}(x)$.

5. $f(x)=6x$. Find $f^{-1}(x)$.

6. $f(x)=x-2$, $g(x)=4x$. Find $(g \circ f)(x)$.

Section 2 — Core practice

7. Find $f^{-1}(x)$ for $f(x) = 3x - 6$.

8. Find $f^{-1}(x)$ for $f(x) = (x + 1)/2$.

9. $f(x)=2x+1$, $g(x)=x^2$. Find $(g \circ f)(x)$.

10. $f(x)=5x-2$, $g(x)=x+3$. Find $(f \circ g)(x)$.

11. Find $f^{-1}(x)$ for $f(x) = x/4 + 2$.

12. $f(x)=x^2$, $g(x)=x-5$. Find $(f \circ g)(7)$.

Section 3 — Challenge & exam-style

13. $f(x) = 2x + 3$. Show that $f^{-1}(f(x)) = x$.

14. $f(x) = (3x - 1)/2$. Find $f^{-1}(x)$.

15. $f(x)=x+2$, $g(x)=x^2$. Find x if $(f \circ g)(x)=11$.

16. If $f(x)=4x-7$, find the value of x for which $f^{-1}(x)=3$.